

Named Entity Recognition (NER)

Named Entity Recognition (NER) is a subtask of Natural Language Processing (NLP) that involves identifying and classifying key information (entities) in text into predefined categories.

- It converts unstructured text into structured information.
- Useful for information retrieval, question answering, chatbots, and analytics.

Purpose:

- Find common entity types in a text.
- Understand the who, what, when, where, and how much in text.

Common Named Entity Types with Examples

Entity Type	Description	Example
PERSON	Names of people	Ravi Kumar, Dr. Joe Smith
ORGANIZATION (ORG)	Names of companies, institutions, organizations	Infosys, Indian Institute of Technology, State Bank of India
LOCATION (LOC)	Cities, countries, rivers, landmarks	India, New York, Mount Everest
DATE	Dates mentioned in text	15th January 2026, March 2025
TIME	Time references	10:30 AM, 22:45
MONEY	Monetary values	\$2000, ₹50,000, €1,500
PERCENT	Percentage values	25%, 12.5%
PRODUCT	Names of products	iPhone, Tesla Model S
EVENT	Names of events, festivals, conferences	Olympics 2024, Diwali, CES 2026

How NER Works

1. Text Preprocessing:
Tokenization, stopword removal, sentence splitting
2. Feature Extraction:
Capitalization, word shape, POS tags, context
3. Entity Recognition:

Using rules, dictionaries, or ML models

4. Classification:

Assign each entity to its predefined type

Example

Text:

"Dr. Joe Smith of Infosys visited New York on 15th January 2026 to attend CES 2026."

NER Output:

- Dr. Joe Smith → Person
- Infosys → Organization
- New York → Location
- 15th January 2026 → Date
- CES 2026 → Event

NER is a fundamental step in NLP that extracts actionable information from text, making it easier for systems to understand and use text intelligently.

Named Entity Recognition (NER) Rules

Rule 1: Person Name (Consecutive Capitalized Words)

- **Pattern:** Consecutive capitalized words, sometimes preceded by capitalized titles.
- **Example:**

Ravi Kumar → Person

Dr. A.P.J. Abdul Kalam → Person

Rule 2: Organization (Keywords)

- **Pattern:** Words followed by keywords like **Ltd, Limited, University, Institute, Org., Company.**
- **Example:**

Infosys → Organization

Indian Institute of Technology → Organization

Rule 3: Location (Known Places)

- **Pattern:** Words matching known countries, states, cities, or landmarks.

- **Example:**

India → Location

Mount Everest → Location

New York City → Location

Rule 4: Money (Currency Symbol + Number)

- **Pattern:** Currency symbol followed by a number.

- **Example:**

\$2000 → Money

₹50,000 → Money

€1,500 → Money

Rule 5: Percentage (Number + % Symbol)

- **Pattern:** Numbers followed by a percentage symbol (%)

- **Example:**

31% → Percent

12.5% → Percent

Rule 6: Organization (Capitalized Phrases Ending with Bank/Corporation/etc.)

- **Pattern:** Capitalized phrase ending with **Bank, Corporation, Company, Services.**

- **Example:**

- State Bank of India → Organization

Reliance Industries → Organization

Rule 7: Person (Titles Followed by Name)

- **Pattern:** Titles like **Mr., Ms., Dr., Prof.** followed by capitalized name.

- **Example:**

M. Ram → Person

Dr. Joe Smith → Person

Rule 8: Date (Preposition + Number/Month)

Pattern: Prepositions like **on, in, since** followed by a number or month often indicate a date.

Example:

On 15th January 2026 → Date

Since March 2025 → Date

Rule 9: Time

- **Pattern:** Numbers followed by AM/PM or 24-hour format.
- **Example:**

10:30 AM → Time

22:45 → Time

Rule 10: Acronyms / Abbreviations

- **Pattern:** All capital letters or letters separated by dots.
- **Example:**

NASA → Organization

U.S.A → Location

Rule 11: PRODUCT

- **Pattern:** Words followed by keywords like phone, mobile, laptop, tablet, watch, camera, or alphanumeric model numbers.
- **Example:** iPhone 14, Dell Inspiron 15, Samsung Galaxy S21

Rule 12: EVENT

- **Pattern:** Capitalized multi-word phrases or keywords like festival, conference, summit, ceremony, workshop, hackathon, day.
- **Example:** Independence Day, CES 2026, Science Workshop

Named Entity Recognition (NER) – BIO Tagging

Definition:

NER identifies and classifies entities in text. To **label each word**, we use the **BIO tagging scheme**:

- **B-XXX** → Beginning of an entity of type XXX
- **I-XXX** → Inside the entity of type XXX
- **O** → Outside any entity

Common Entity Types

- **PERSON** → Names of people
- **ORG / ORGANIZATION** → Companies, Institutions
- **LOC / LOCATION** → Cities, Countries, Landmarks
- **DATE** → Dates
- **TIME** → Time references
- **MONEY** → Currency values
- **PERCENT** → Percentages
- **PRODUCT** → Products, devices, gadgets
- **EVENT** → Named events, festivals, conferences

NER Tagging Examples**Example 1: PERSON, ORG, LOC**

Text: *Mr. Rajesh Verma works at Tata Consultancy Services in Mumbai.*

Step 1: Tokenize the Text

First, split the sentence into individual words (tokens):

["Mr.", "Rajesh", "Verma", "works", "at", "Tata", "Consultancy", "Services", "in", "Mumbai"]

Step 2: Identify Named Entities

Look for **words or phrases that represent entities**:

- **Mr. Rajesh Verma** → Person
- **Tata Consultancy Services** → Organization
- **Mumbai** → Location

Everything else (verbs, prepositions, etc.) → **not an entity**

Step 3: Apply BIO Rules

BIO tagging rules:

- **B-XXX** → Beginning of an entity type XXX
- **I-XXX** → Inside the entity type XXX (following B-XXX)
- **O** → Outside any entity

Step 4: Tag Each Word

Word	Reasoning	Tag
Mr.	Title indicating a person, first word of a person name	B-PERSON
Rajesh	Continuation of the person's name	I-PERSON
Verma	Continuation of the person's name	I-PERSON
works	Verb, not an entity	O
at	Preposition, not an entity	O
Tata	First word of an organization name	B-ORG
Consultancy	Continuation of organization name	I-ORG
Services	Continuation of organization name	I-ORG
in	Preposition, not an entity	O
Mumbai	Location, first and only word in the location entity	B-LOC

Step 5: Final BIO Tags

Sentence with tags:

["Mr." → B-PERSON, "Rajesh" → I-PERSON, "Verma" → I-PERSON, "works" → O, "at" → O, "Tata" → B-ORG, "Consultancy" → I-ORG, "Services" → I-ORG, "in" → O, "Mumbai" → B-LOC]

Example 2: DATE, EVENT

Text: *Independence Day is celebrated on 15 August.*

Tagging:

Word	Tag
Independence	B-EVENT
Day	I-EVENT
is	O
celebrated	O
on	O
15	B-DATE
August	I-DATE

Example 3: PRODUCT and MONEY

Text: *The mobile phone iPhone 14 costs \$1,200.*

Tagging:

Word	Tag
The	O
mobile	B-PRODUCT
phone	I-PRODUCT
iPhone	I-PRODUCT
14	I-PRODUCT
costs	O
\$1,200	B-MONEY

Example 4: PERSON, ORG, EVENT, DATE

Text: *Dr. A.P.J. Abdul Kalam inaugurated the National Science Conference at Anna University on 12 March 2010.*

Tagging:

Word	Tag
Dr.	B-PERSON
A.P.J.	I-PERSON
Abdul	I-PERSON
Kalam	I-PERSON
inaugurated	O
the	O
National	B-EVENT
Science	I-EVENT
Conference	I-EVENT
at	O
Anna	B-ORG
University	I-ORG
on	O
12	B-DATE
March	I-DATE
2010	I-DATE

Example 5: PERSON, LOC

Text: *Mr. Suresh Kumar lives in Coimbatore.*

Tagging:

Word	Tag
Mr.	B-PERSON
Suresh	I-PERSON
Kumar	I-PERSON

Word	Tag
lives	O
in	O
Coimbatore	B-LOC